POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Artificial Intelligence

Semester: Fall

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Define intelligence. With reference to Turning Test can we infer that machine possesses Intelligence, justify your answer. 7
   b) Define State Space in problem solving. Using constraint satisfaction problem, solve the following crypto-arithmetic problem. 8
   CROSS+ROADS=DANGER
2. a) What are the different types of problems? How can you say that a problem is well defined or not. 8
   b) In what condition the blind search is preferred? Explain Breadth first search with it's algorithms. 7
3. a) Compare and contrast greedy search and A* search with a suitable example. 8
   b) Define Semantic Network. Draw semantic network of following clauses: 7
   Subset_of(Human, Mammal), Subset_of(Male, Human),
   Subset_of(Female, Human), Has_Mother(Human, Female),
   Member_of(Mary, Female), Member_of(John, Male),
   Husband_of(John, Mary).
4. a) Assume the following facts: 8
   - Steve only likes easy courses
   - Science courses are hard
   - All the course in the basket weaving department are easy
   - BK301 is a basket weaving course

   Use resolution to answer the question," What course would Steve like?"
   b) Using Truth Table, Prove the $P \leftrightarrow Q$ is equivalent to $(P \rightarrow Q) \wedge (Q \rightarrow$ 7

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P)

5. a) Define clustering and K-means clustering. Discuss briefly about fuzzy learning and its importance. 7

b) How does rule based deduction system work. Compare it with the working of Bayes' theorem. 8

6. a) What do you understand by neural network? Explain about multilayer perception with proper diagrams. 7

b) What is neutral network? Define perceptron. How is winning node chosen in Kohonen network, illustrate with following dataset: 8

$$X = \begin{bmatrix} 0.52 \\ 0.12 \end{bmatrix} \quad W_1 = \begin{bmatrix} 0.27 \\ 0.81 \end{bmatrix} \quad W_2 = \begin{bmatrix} 0.42 \\ 0.70 \end{bmatrix} \quad W_3 = \begin{bmatrix} 0.43 \\ 0.21 \end{bmatrix}$$

Where,

X: two dimensional input vector presented in 3-dimentional Kohonen Network

$$W_i = weight\ vectors, i = 1,2,3$$

7. Write short notes on: (**Any two**) 2×

a) Alpha Beta procedure.

b) Tautology.

c) Frames

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